



Operating, Logging & Uploading

Session 3 — The On-Air Workflow

TavaOne Education · W4GGJ



Today's Session

- Band and mode selection — when to be where
- CQ POTA procedure and exchange
- Self-spotting on pota.app
- FT8 from the field — why it matters
- HAMRS logging — walkthrough from create to ADIF
- Log upload to pota.app — full workflow
- **Simulated activation — 10 QSOs start to upload**



Band & Mode Selection

When to be where, and why



Band Strategy for Florida Activations

- 40m — the workhorse. Morning/afternoon: regional (0–500 mi). Night: 500–2500 mi.
- 20m — midday through mid-afternoon. Best for 1,000–4,000 mi hunters.
- 15m / 10m — when solar flux is high. Check before going. Often excellent in Solar Cycle 25.
- 80m — evenings, high noise, short range. Backup when 40m skips over your area.
- VHF FM — 147.555 simplex. Any time. Local hunters, quick opener.

- Pre-activation check:
 - PSKReporter.info — see what bands are open in your area
 - DX Maps app — real-time propagation map
 - POTA Spots page — what bands active activators are using right now



Mode Tradeoffs



Phone (SSB)

- 20–60 contacts/hour in good conditions
- No extra gear needed
- Easy to manage a pileup verbally
- Easiest way to explain you're in a park
- Signal must be strong enough for voice
- Most fun — you talk to real people



FT8/FT4

- 30–80 contacts/hour, marginal conditions OK
- Needs laptop/tablet + audio interface
- 15 dB better than SSB weak-signal
- Excellent for days when SSB is tough
- High TX duty cycle — watch power output
- Works while you're sipping coffee



CQ POTA — The On-Air Workflow

From first call to last contact



Finding a Clear Frequency

- Before you transmit:
 - Listen for 30–60 seconds on the frequency
 - Ask "Is this frequency in use?" (CW: "QRL?")
 - Wait 5 seconds — if no response, proceed
- Common POTA frequency ranges (not exclusive):
 - 40m SSB: 7.200–7.290 MHz
 - 20m SSB: 14.150–14.300 MHz
 - 15m SSB: 21.200–21.400 MHz
 - 40m CW: 7.025–7.060 MHz
- Avoid: known repeater inputs, other spotted activators, beacon frequencies
- If the band is crowded: go up 5 kHz and try again



The CQ POTA Exchange

CQ POTA, CQ POTA — this is Whiskey Four Golf Golf Juliet, W4GGJ
activating K-2119, Honeymoon Island State Park — W4GGJ listening

Exchange:

HUNTER: W4GGJ de K1ABC, 59 in Massachusetts

W4GGJ: K1ABC, you're 57, thanks for the Parks, 73

HUNTER: Thanks! 73

W4GGJ: QRZ? W4GGJ ← go immediately to next caller



Managing a Pileup

- When spotting works, multiple hunters call simultaneously
- Rules for working a pileup:
 - 1. Pick ONE callsign you heard clearly
 - 2. Call them back: "W1ABC, you're 57, thanks for the Parks"
 - 3. QSL briefly, then: "QRZ? W4GGJ" — go immediately
 - 4. Never stop calling — dead air loses hunters
 - 5. Never chat during the run
- If you can't pull a full callsign:
 - "The station ending in Lima?"
 - "Partial copy — last two characters?"
- If QRM builds: move 5 kHz, re-spot new frequency



Self-Spotting

Getting hunters on your frequency



Why Spotting Is Non-Negotiable

- Without a spot on pota.app, hunters don't know you're active
- A good spot can produce 5–10 callers within 60 seconds
- How to self-spot:
 - pota.app → Activator Tools → Spot Yourself
 - Enter: callsign, park reference, frequency, mode
 - Optional comment: "40m SSB running, come get it"
- Spot accurately:
 - Frequency must match what you're actually on (not 7.245 when you're on 7.247)
 - If you QSY, re-spot immediately
- Also visible to hunters via:
 - DX cluster spots ([DXsummit.fi](https://dxsummit.fi))



FT8 from the Field

The secret weapon for marginal conditions



Why Learn FT8 for POTA

- FT8 operates 15 dB below the noise floor — far beyond voice range
- A band that sounds "dead" may have active FT8 contacts happening on it
- Practical impact:
 - SSB might get you 3 contacts on a bad band day
 - FT8 gets you 10 contacts on that same day
- Standard field setup:
 - WSJT-X software (free, all platforms)
 - Digirig (\$50) + cable for most radios
 - OR IC-705 built-in USB (no extra hardware)
- POTA FT8 frequencies:
 - 40m: 7.074 MHz (dial USB)



FT8 Duty Cycle Warning

- FT8 is 100% duty cycle — full RF output for 13.5 of every 15 seconds
- Implications:
 - PA runs hot — reduce power for FT8 vs SSB
 - Rule of thumb: 50% of max power rating for FT8
 - FT-891: don't exceed 50W for FT8
 - IC-705: 5W is already conservative — fine at 5W
- Battery impact:
 - FT8 at 50W $\approx$ constant 12A draw (no RX recovery)
 - Plan shorter runtime or bigger battery for FT8 sessions
- Contacts per minute:
 - Once spotted and running: 3–6 contacts/minute is achievable



Field Logging

Paper, HAMRS, and WSJT-X



HAMRS — Full Walkthrough

- 1. Open HAMRS → tap + (new activation)
- 2. Select POTA template
- 3. Enter your callsign and park reference (K-2119)
- 4. Set default band and mode
- 5. Tap Start Activation

- Logging a QSO:
 - Type callsign in CALL field
 - Date and time auto-populate in UTC
 - Adjust band/mode if changed
 - Enter RST sent (59 for SSB default)
 - Tap Log QSO

- Exporting ADIF:



What ADIF Looks Like

```
<CALL:5>K1ABC <QSO_DATE:8>20260919 <TIME_ON:4>1532  
<BAND:3>40M <MODE:3>SSB <RST_SENT:2>59  
<SIG:4>POTA <SIG_INFO:6>K-2119 <EOR>
```

Critical fields:

- CALL — their callsign
- QSO_DATE — YYYYMMDD in UTC
- TIME_ON — HHMM in UTC
- BAND — 40M, 20M, 15M, etc.
- **SIG — must be exactly "POTA"**
- SIG_INFO — your park reference (K-2119)



Uploading to pota.app

- Log in to pota.app
- My Activations → Upload Log
- Select your ADIF file
- Preview the upload — check park ref and contact count
- Click Submit
- Processing: usually instant
- Common errors:
 - "0 valid POTA contacts" → missing SIG/SIG_INFO fields
 - "Invalid park reference" → typo; must include dash (K-2119)
 - "Duplicate contact" → normal if re-uploading; duplicates silently rejected
- After upload:
 - Activation appears on your profile



Awards & Milestones

What you're chasing



POTA Award Structure

- Activator awards — based on valid activation count:
 - 1, 10, 25, 50, 100, 250, 500, 1000 activations
- Park awards — unique parks activated:
 - Separate from activation count
 - Rare to repeat the same park a lot early on — explore
- Hunter awards — unique parks worked as a hunter:
 - 1, 10, 25, 50, 100, 250, 500, 1000 parks
 - Tracked automatically from uploaded hunter logs
- P2P awards — park-to-park contacts:
 - When you're activating and work another activator
 - Both get P2P credit — separate award track

From CQ to Upload

- Instructor calls CQ POTA on club station (mock park)
- Students call back — take turns at the mic
- All students log simultaneously in HAMRS on phones
- Paper log backup running simultaneously
- Group reaches 10 QSOs together
- Export ADIF from HAMRS — open in text editor
- **Upload to [pota.app](#) test account — watch it go live**



KNOWLEDGE CHECK

The standard FT8 calling frequency on 20 meters is:

A. 14.000 MHz (CW calling frequency)

B. 14.074 MHz (FT8 calling frequency)

C. 14.225 MHz (SSB general)



KNOWLEDGE CHECK

You upload your ADIF and pota.app says "0 valid POTA contacts found." The most likely cause is:

A. The contacts were on the wrong date

B. The SIG and SIG_INFO fields are missing from the ADIF

C. You didn't wait 24 hours before uploading



KNOWLEDGE CHECK

After calling CQ POTA for 10 minutes with no response, what should you try FIRST?

A. Give up and go home

B. Check if you self-spotted on pota.app with the correct frequency

C. Move to 80m immediately



Session 4 — Field Activation Capstone

The Real Thing

- Local POTA park — real activation, real contacts
- Team setup: everyone assembles the station
- Every student operates — at least 1 contact at the mic
- Every student logs in HAMRS simultaneously
- Group self-spots on pota.app
- Upload log live — watch activation go live on pota.app
- **TavaOne Education POTA Activator Certificate issued**